

PASSWORD CRACKING REPORT

JOHN THE RIPPER & NETWORKWALKS PASSWORD CRACKING

W3-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B083-Networkwalks
Modules completed	W3-PM1 (Password Cracking with JTR) W3-PM2 (Networkwalks Password Cracking Tools)
Client/Target	Protected PDF files supplied as part of the Networkwalks educational project
Permission secured from client?	Yes — activities were performed within the assigned educational scope using the supplied lab materials.
Phases covered	Phase 1: Hash Extraction & Preparation Phase 2: Password Recovery Phase 3: Credential Verification & Evidence Documentation

1. Liability Disclaimer

I have performed these activities only within the assigned educational cybersecurity scope and against the files provided or authorized for this training exercise. All materials are for education and research purposes only. Do not use anything from this report to break the law. The instructor, the authors and Networkwalks are not responsible for misuse of this knowledge. Every action taken outside an authorized scope is the responsibility of the person performing it. Unauthorized access can be a criminal offence even when no damage is caused.

2. Introduction

This report covers the Week 3 practical activities completed as part of the Cybersecurity & Ethical Hacking program at Networkwalks. The work focuses on password recovery from protected PDF files using John the Ripper (JTR) with Johnny GUI and the Networkwalks Hash Calculator and Password Cracker. The practical work was completed in a controlled educational environment and is supported by the evidence screenshots included in this report.

3. Tools Used

John the Ripper, Johnny GUI, Networkwalks Hash Calculator, Networkwalks Password Cracker, and the protected PDF challenge files supplied for the lab.

4. Activities Performed

4.1 W3-PM1 — Password Cracking with John the Ripper (JTR)

The JTR workflow covered installation/build verification, Johnny configuration, PDF1 hash extraction, hash loading, password recovery, and final PDF verification.

```
Windows PowerShell
PS C:\Users\Faizan\Downloads\john-1.9.0-jumbo-1-win64\run> .\john.exe --list=build-info
Version: 1.9.0-jumbo-1
Build: cygwin 64-bit x86_64 AVX2 AC OMP
SIMD: AVX2, interleaving: MD4:3 MD5:3 SHA1:1 SHA256:1 SHA512:1
CPU tests: AVX2
CPU fallback binary: john-xop
OMP fallback binary: john-avx2-non-omp
$JOHN is /run/
Format interface version: 14
Max. number of reported tunable costs: 4
Rec file version: REC4
Charset file version: CHR3
CHARSET_MIN: 1 (0x01)
CHARSET_MAX: 255 (0xff)
CHARSET_LENGTH: 24
SALT_HASH_SIZE: 1048576
SINGLE_IDX_MAX: 2147483648
SINGLE_BUF_MAX: 4294967295
Effective limit: Number of salts vs. SingleMaxBufferSize
Max. Markov mode level: 400
Max. Markov mode password length: 30
gcc version: 7.4.0
OpenCL headers version: 2.2
Crypto library: OpenSSL
OpenSSL library version: 01010102f
OpenSSL 1.1.1b 26 Feb 2019
GMP library version: 6.1.2
File locking: fcntl()
fseek(): fseek
ftell(): ftell
fopen(): fopen
memmem(): System's
PS C:\Users\Faizan\Downloads\john-1.9.0-jumbo-1-win64\run>
```

Figure 1 — JTR build and version information.

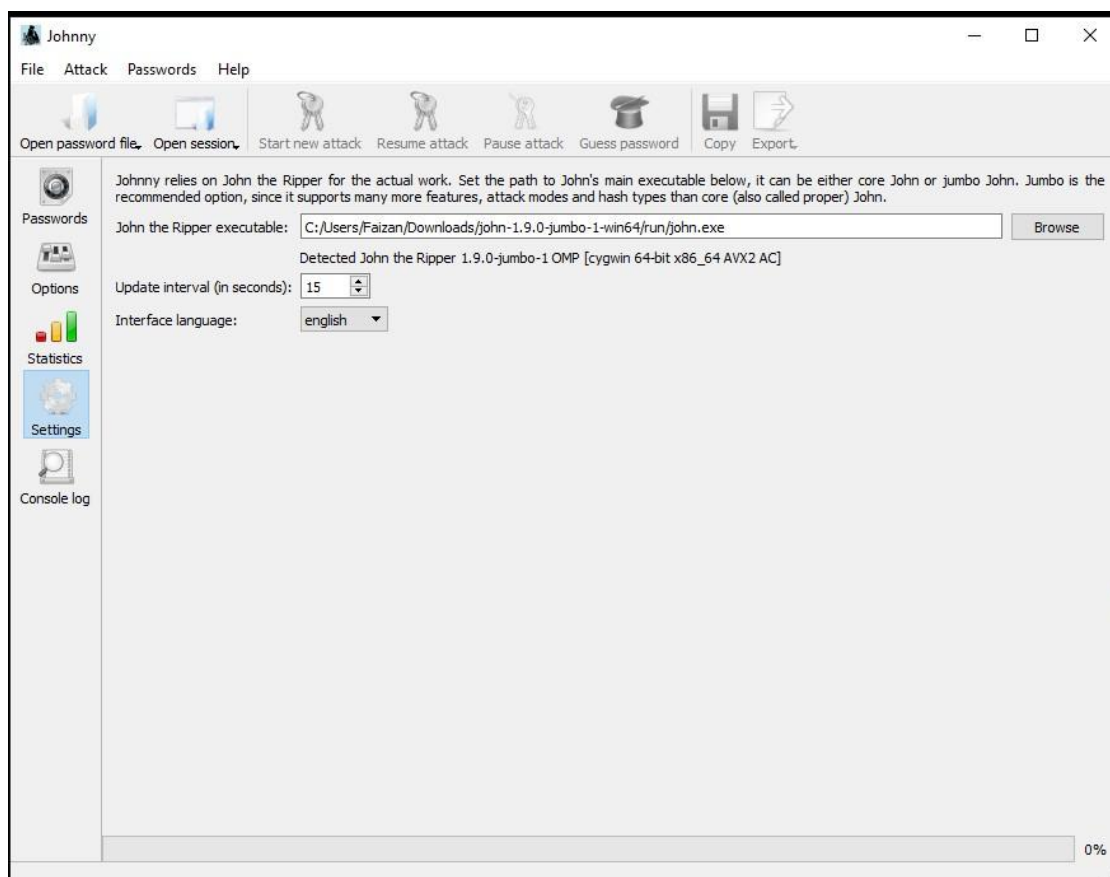


Figure 2 — Johnny configuration showing the JTR executable.

Upload an encrypted PDF

The converter extracts the format used by **pdf2john** and **pdf2hashcat**. Upload a file only when you have permission to test it; the source is deleted after extraction.

Select a PDF file

My Locked PDF1.pdf

Choose file

Files are processed for extraction and deleted after extraction. Maximum upload size: 200 MB.

Extraction complete. Review the output before using it in an authorized test.

Extracted hash or conversion output

```
$pdf$4*4*128*-1028*1*16*ca7f72f11459cba469f1005a8765ed51*32*f32d8fa1bfbe  
2648226dfc39f7909ea0021446990b9e4114071a4d9104984c1*32*9322f50c29569712  
067a775264635e4954ccb1b99e209d664984054ffad30a6a
```

Figure 3 — Extracted hash for PDF1.

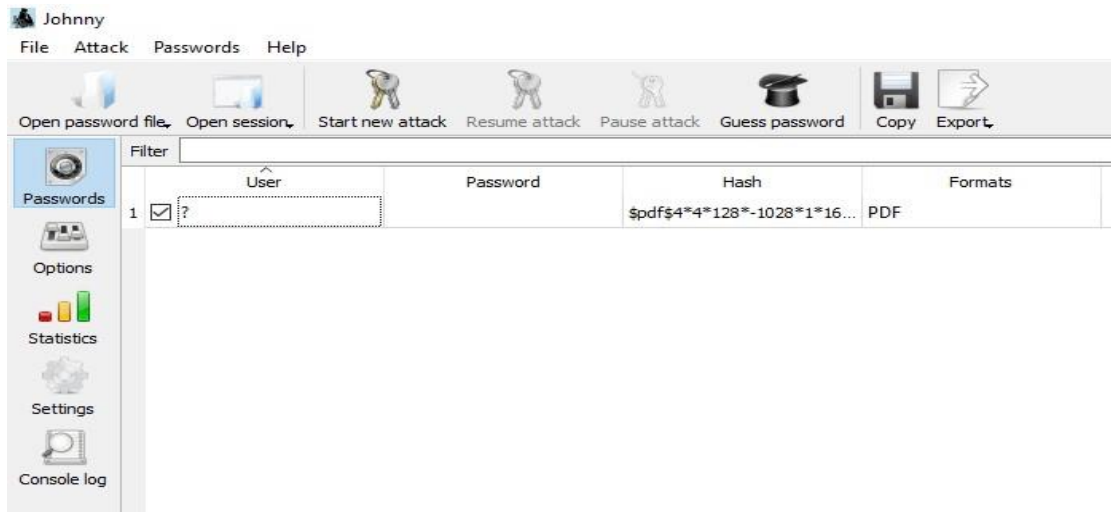


Figure 4 — PDF1 hash loaded into Johnny.



Figure 5 — Successful PDF1 password recovery.



Figure 6 — PDF1 successfully verified.

4.2 W3-PM2 — Password Cracking with Networkwalks Tools

The Networkwalks workflow covered PDF2 hash extraction, password recovery, flag capture, and final PDF verification.

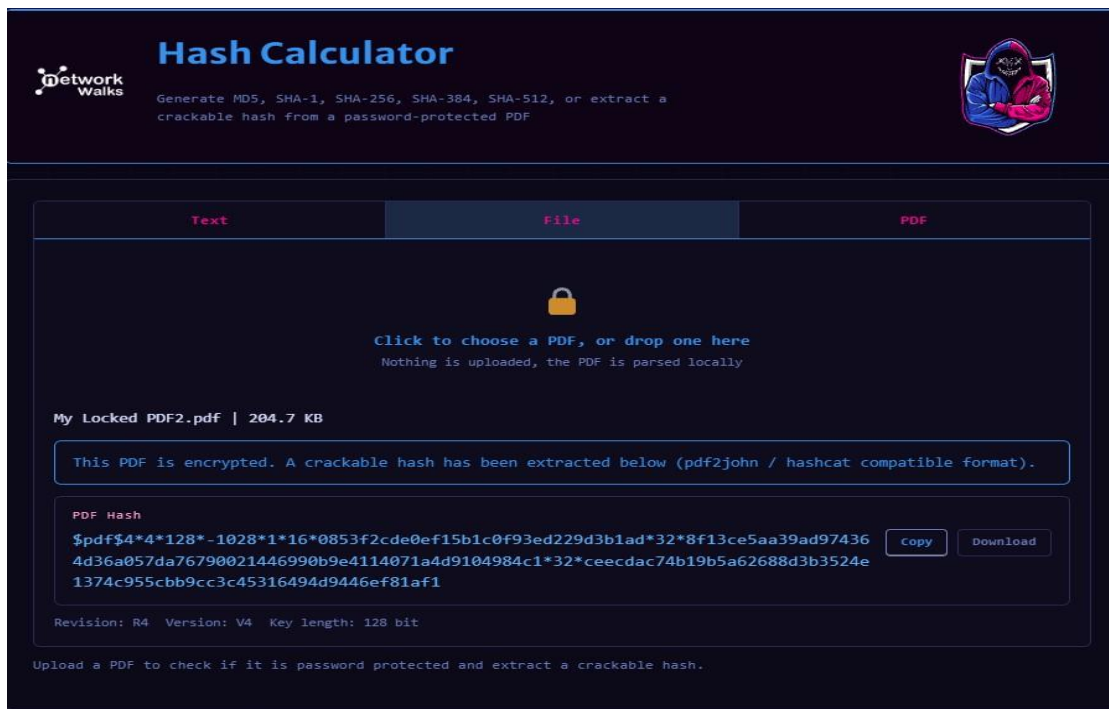


Figure 7 — PDF2 hash extracted using the Networkwalks Hash Calculator.

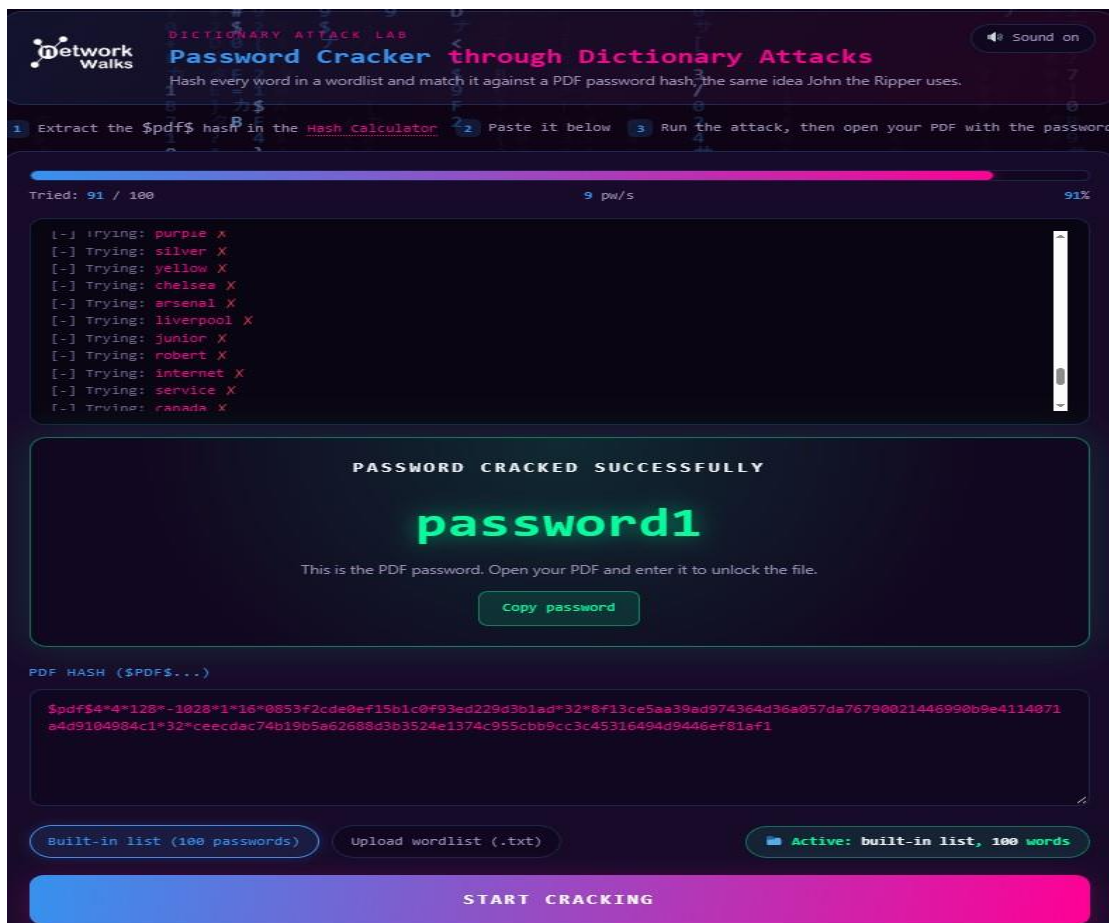


Figure 8 — Successful PDF2 password-cracking result and flag evidence.

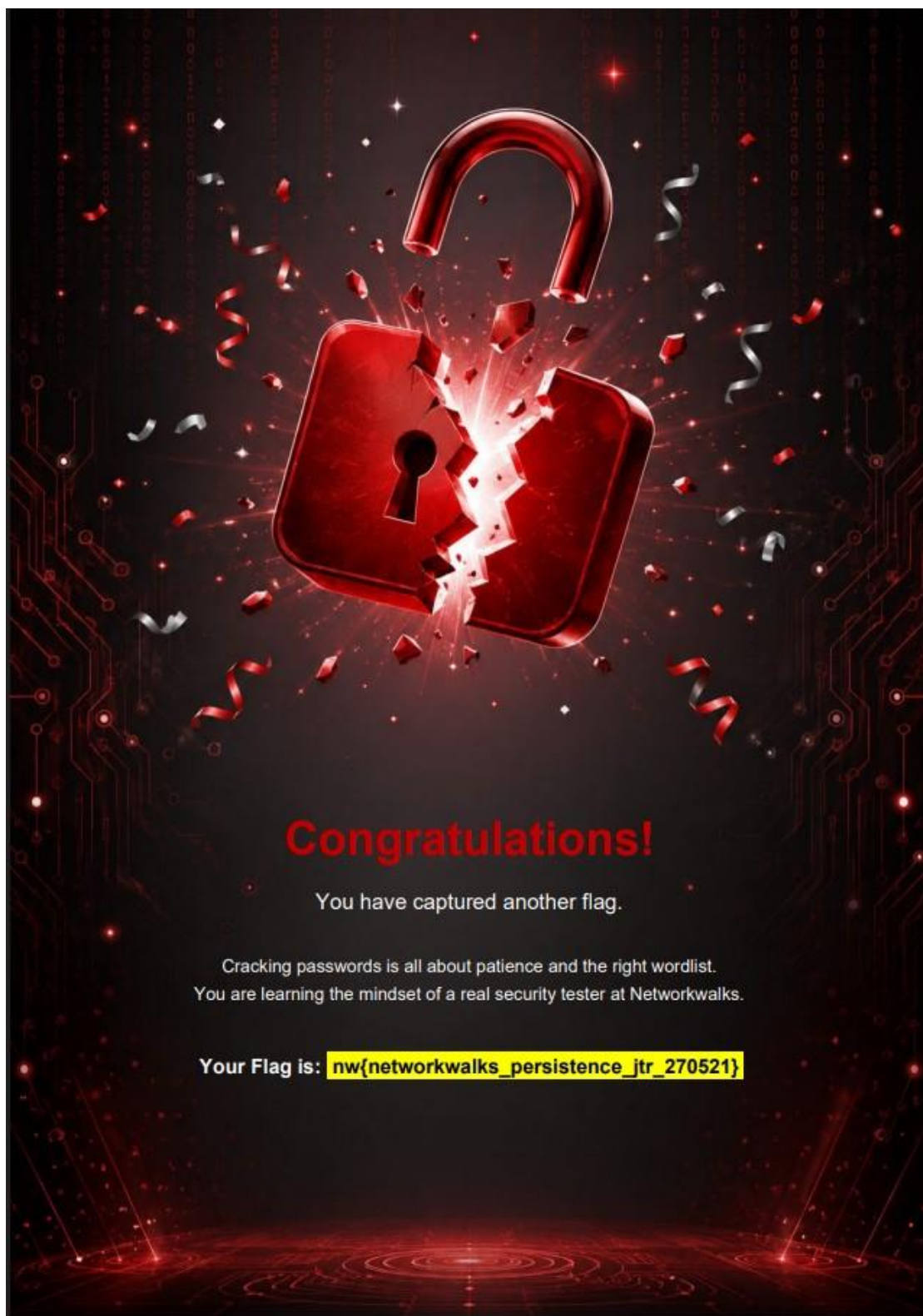


Figure 9 — PDF2 successfully verified.

5. Overall Results

W3-PM1 was completed successfully using JTR/Johnny for PDF1. W3-PM2 was completed successfully using Networkwalks tools for PDF2, including successful password recovery, flag capture and verification.

6. Conclusion

The Week 3 project provided practical experience with PDF password hashes, password-recovery workflows, verification and cybersecurity evidence documentation. All activities were completed within the authorized educational scope.

7. GitHub Repository

<https://github.com/faizan-manazir/Neworkwalks-B083-Wk3-Password-Cracking>

8. Ethical Use Statement

The techniques documented in this report are intended for authorized security testing, education and research only. Password cracking against files, systems or accounts without explicit permission is unauthorized and may be illegal.